

1) Username

```
bindu@Bindusree: ~  
bindu@Bindusree:~$ nano username.c  
bindu@Bindusree:~$ gcc username.c -o username  
bindu@Bindusree:~$ ./username  
Enter your name:Bindu  
Welcome Bindu  
bindu@Bindusree:~$ |
```

```
GNU nano 8.7.1  
#include<stdio.h>  
int main()  
{  
char name[30];  
printf("Enter your name:");  
scanf("%s",name);  
printf("Welcome %s\n",name);  
return 0;  
}
```

2)

```
bindu@Bindusree: ~  
GNU nano 8.7.1  
#include<stdio.h>  
int main()  
{  
printf("Welcome to operating system\n");  
return 0;  
}
```

```
bindu@Bindusree:~$ nano username.c  
bindu@Bindusree:~$ nano welcome.c  
bindu@Bindusree:~$ gcc welcome.c -o welcome  
bindu@Bindusree:~$ ./welcome  
Welcome to operating system  
bindu@Bindusree:~$ |
```

3)precessid

```
bindu@Bindusree:~$ nano processid.c  
bindu@Bindusree:~$ gcc processid.c -o processid  
bindu@Bindusree:~$ ./processid  
Process ID=767  
bindu@Bindusree:~$ |
```

```
bindu@Bindusree: ~  
GNU nano 8.7.1 processid.c  
#include<stdio.h>  
#include<unistd.h>  
int main()  
{  
printf("Process ID=%d\n",getpid());  
return 0;  
}
```

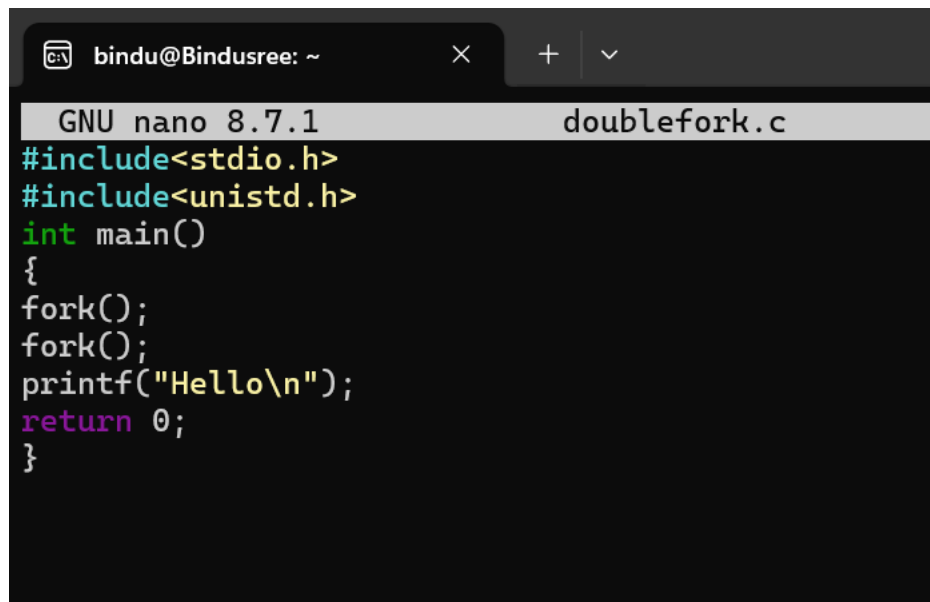
4)fork

```
bindu@Bindusree: ~  
bindu@Bindusree:~$ nano fork.c  
bindu@Bindusree:~$ gcc fork.c -o fork  
bindu@Bindusree:~$ ./fork  
Hello  
Hello  
bindu@Bindusree:~$
```

```
bindu@Bindusree: ~  
GNU nano 8.7.1  
#include<stdio.h>  
#include<unistd.h>  
int main()  
{  
fork();  
printf("Hello\n");  
return 0;  
}
```

5)Double fork

```
bindu@Bindusree:~$ nano doublefork.c
bindu@Bindusree:~$ gcc doublefork.c -o doublefork
bindu@Bindusree:~$ ./doublefork
Hello
Hello
Hello
Hello
bindu@Bindusree:~$ |
```



The screenshot shows a terminal window with a tab labeled 'bindu@Bindusree: ~'. The window contains the GNU nano 8.7.1 editor editing the file 'doublefork.c'. The code is as follows:

```
GNU nano 8.7.1 doublefork.c
#include<stdio.h>
#include<unistd.h>
int main()
{
fork();
fork();
printf("Hello\n");
return 0;
}
```

6) Parentchild

```
bindu@Bindusree:~$ nano parentchild.c
bindu@Bindusree:~$ gcc parentchild.c -o parentchild
bindu@Bindusree:~$ ./parentchils
-bash: ./parentchils: No such file or directory
bindu@Bindusree:~$ ./parentchild
I am ParentProcess
I am Child Process
bindu@Bindusree:~$ |
```

```
bindu@Bindusree: ~  
GNU nano 8.7.1 parentchild.c  
#include<stdio.h>  
#include<unistd.h>  
int main()  
{  
pid_t pid;  
pid=fork();  
if(pid==0)  
{  
printf("I am Child Process\n");  
}  
else  
{  
printf("I am ParentProcess\n");  
}  
}
```

7)Parent

```
bindu@Bindusree:~$ nano parent.c  
bindu@Bindusree:~$ gcc parent.c -o parent  
bindu@Bindusree:~$ ./parent  
Parent PID = 1170  
Child PID = 1171  
Parent PID= 1171  
bindu@Bindusree:~$
```

```
bindu@Bindusree: ~  
GNU nano 8.7.1 parent.c  
#include<stdio.h>  
#include<unistd.h>  
int main()  
{  
pid_t pid;  
pid=fork();  
if(pid==0)  
{  
printf("Child PID = %d\n",getpid());  
printf("Parent PID= %d\n",getpid());  
}  
else  
{  
printf("Parent PID = %d\n",getpid());  
}  
}
```

8)wait

```
bindu@Bindusree:~$ nano wait.c  
bindu@Bindusree:~$ gcc wait.c -o wait  
bindu@Bindusree:~$ ./wait  
Child Running  
Parent Running After Child  
bindu@Bindusree:~$
```



bindu@Bindusree: ~



GNU nano 8.7.1

wait.c

```
#include<stdio.h>
#include<unistd.h>
#include<sys/wait.h>
int main()
{
    pid_t pid;
    pid=fork();
    if(pid==0)
    {
        printf("Child Running\n");
    }
    else
    {
        wait(NULL);
        printf("Parent Running After Child\n");
    }
    return 0;
}
```